

## POKHARA UNIVERSITY

Level: Bachelor  
Programme: BE  
Course: Calculus II

Semester: Fall

Year : 2024  
Full Marks : 100  
Pass Marks : 45  
Time : 3 hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

1. a) Evaluate:  $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$  by changing the order of integration. 5
  - b) Evaluate the integrals:  $\iiint_V (x^2 + y^2 + z^2) dx dy dz$  taken over the volume of the sphere  $x^2 + y^2 + z^2 = 1$ . 5
  - c) Find the volume of the solid whose base is the region in the xy-plane that is bounded by the parabola  $y = 4 - x^2$  and the line  $y = 3x$ , while the top of the solid is bounded by the plane  $z = x + 4$ . 5
  2. a) Solve the differential equation:  $y'' + 9y = 0$  by using power series method. 7
  - b) Define Legendre differential equation. Also, derive the solution of Legendre differential equation. 8
- OR**
- If  $J_n(x)$  represents the Bessel's function of order  $n$ , then show that:
- i.  $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$  4
  - ii.  $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$  4
3. a) Solve the following differential equation by using Laplace transform  $y'' + 2y' - 3y = 6e^{-2t}$ ,  $y(0) = 2$ ,  $y'(0) = -14$ . 7
  - b) i) State second shifting property for Laplace transform and using this, find the Laplace transform of  $e^{-3t} u(t - 3)$ . 4
  - ii) Find the inverse Laplace transform of  $\frac{\pi}{(s-3)^2 + \pi^2}$ . 4
4. a) Find the directional derivative of  $f = x^2 yz + 4xz^2$  at  $(1, -2, 1)$  in the direction  $2\vec{i} - \vec{j} - 2\vec{k}$ . 5

- b) Show that the vector  $\vec{F} = (x^2 - yz, y^2 - xz, z^2 - xy)$  is conservative vector field and find the scalar potential function  $\phi$  such that  $\vec{F} = \nabla\phi$ . 5
- c) Find the work done by the force  $\vec{F} = (3x^2, 2xz - y, z)$  along the curve  $x = t, y = \frac{t^2}{4}, z = \frac{3t^2}{8}$  from  $(0, 0, 0)$  to  $(2, 1, 3)$ . 5
5. a) Evaluate the integral by using Green's theorem 7
- $\oint_C [(3x^2 + y) dx + 4y^2 dy]$  where  $C$ : the boundary of the triangle with vertices  $(0, 0), (1, 0), (0, 2)$ ; oriented in counterclockwise direction.
- b) State Stoke's Theorem. Evaluate  $\oint_C (\vec{F} \cdot d\vec{r})$  by using Stoke's theorem, where  $\vec{F} = (y, xz^3, -zy^3)$  and  $C: x^2 + y^2 = 4, z = 3$ . 8
- OR**
- Evaluate:  $\iint_S \vec{F} \cdot \vec{n} dA$ , where  $\vec{F} = (18z, -12, 3y)$  and  $S$  is the surface of plane  $2x + 3y + 6z = 12$  in the first octant.
6. a) Find the Fourier series of  $f(x) = x + |x|, -\pi < x < \pi$  and show that  $\frac{\pi^2}{8} = 1 + \frac{1}{9} + \frac{1}{25} + \dots$ . 7
- b) Find the Fourier sine and cosine series for the function  $f(x) = x^2$  in the interval  $0 < x < L$ . 8
7. Write short notes on: (Any two) 2×5
- a) Find the general solution of the partial differential equation  $u_x + u_y = u$ .
- b) Derive the one-dimensional traffic flow model using conservation law.
- c) A particle moves along the curve  $x = t^3 + 1, y = t^2, z = 2t + 5$ . Find the velocity and acceleration at  $t = 1$ .

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1. 5
  - a) Evaluate  $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2) dx dy$  by changing into polar integration. 5
  - b) Evaluate:  $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$  5
  - c) Find the volume in the first octant bounded by the coordinate planes, the cylinder  $x^2 + y^2 = 4$  and  $z + y = 3$ . 5
2. 8
  - a) i) Find the Laplace Transform of:  $\frac{\sin^2 t}{t}$  8  
ii) Find the inverse Laplace Transform of:  $\frac{w}{s^2(s^2+w^2)}$
  - b) Solve the initial value problem:  $y'' + 2y' - 3y = 6e^{-2t}$  where,  $y(0)=2$ ,  $y'(0)=-14$  using Laplace Transform. 7
3. 7
  - a) Solve  $y'' + y = 0$  by using power series method. 7
  - b) Derive Rodrigue's formula for Legendre polynomials. 8

OR

Define Bessel's differential equation of order  $n$ . Find its complete solution.
4. 5
  - a) Find the directional derivatives of  $f = xy^2 + yz^3$  at  $P(2, -1, 1)$  along the direction of the normal to the surface  $x \log z - y^2 + 4 = 0$  at  $(-1, 2, 1)$ . 5
  - b) If  $\phi = \log(x^2 + y^2 + z^2)$ , find  $\text{div}(\text{grad } \phi)$  and  $\text{curl}(\text{grad } \phi)$ . 5
  - c) Show that the value under integral sign is exact and then evaluate integral, 5

(1,1,1)

$$\int [yz \sinh(xz) dx + \cosh(xz) dy + xy \sinh(xz) dz].$$

(0,2,3)

5. a) Let  $S$  be the hemi-sphere given by  $z = \sqrt{9 - x^2 - y^2}$  and  $\vec{F} = (3x, 3y, z)$ . Find the flux of  $\vec{F}$  through  $S$ .
- b) State Gauss divergence theorem. Evaluate  $\iint_S \vec{F} \cdot \vec{n} dA$  where  $\vec{F} = (y^3, x^3, z^3)$  and  $S$  is the cylinder  $x^2 + 4y^2 = 1$   $0 < z < 5$ ,  $x > 0$ ,  $y > 0$  by using it. 8

OR

State Stoke's theorem. Evaluate  $\oint_C \vec{F} \cdot d\vec{r}$  where  $\vec{F} = (y^3, 0, x^3)$  and  $C$  is the boundary of the triangle with vertices are given by the vectors  $(1, 0, 0)$ ,  $(0, 1, 0)$ ,  $(0, 0, 1)$ .

6. a) Find the fourier series of  $f(x) = x + |x|$  for  $-\pi < x < \pi$ . 7
- b) Find the fourier cosine series of  $f(x) = x \sin x$ ,  $0 < x < \pi$  then show that  $\frac{\pi-2}{4} = \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \frac{1}{7.9} + \dots$ . 8
7. Attempt any two questions. 2×5
- a) Solve:  $u_t + 2uu_x = 0$ ,  $u(x, 0) = e^{-x}$ .
- b) Derive the one-dimensional traffic flow model using conservation law.
- c) Find the angle between the surfaces  $xy^2z - 3x - z^2 = 0$  and  $3x^2 + 2z = 1 + y^2$  at  $(1, -2, 1)$

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*Attempt all the questions.*

1. a) Evaluate  $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2) dx dy$  by changing into polar integration. 5
- b) Evaluate:  $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$  5
- c) Find the volume in the first octant bounded by the coordinate planes, the cylinder  $x^2 + y^2 = 4$  and  $z + y = 3$ . 5
2. a) Solve the differential equation  $y'' + (1 - x^2)y = 0$  by using power series method. 7
- b) Define Legendre's equation. Also derive the solution of Legendre's equation. 8

OR

If  $J_n(x)$  represents the Bessel's function of order  $n$  then show that:

i.  $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x).$

ii.  $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x,$

3. a) State First Shifting Theorem of Laplace transforms. 8
- i. Find the Laplace transform of  $te^{-t} \cosh t$
- ii. Find the inverse Laplace transform of  $\frac{1}{s^2 - 5s + 6}$
- b) Solve the initial value problem:  
 $y'' + 4y' + 3y = e^{-t}, y(0) = y'(0) = 1$  by using Laplace transform. 7
4. a) If  $\Phi = \log(x^2 + y^2 + z^2)$  then find  $\text{div}(\text{grad} \Phi)$ . 5

- b) Find the directional derivative of  $f = x^2yz + 4xz^2$  at  $(1, -2, 1)$  in the direction  $2\vec{i} - \vec{j} - 2\vec{k}$ . 5
- c) Calculate:  $\oint_C \vec{F} \cdot d\vec{r}$  if  $\vec{F} = (y, z, x)$ ,  $C$ ;  $\vec{r} = (t, t^2, t^3)$ , from  $(0, 0, 0)$  and  $(2, 4, 8)$ . 5
5. a) Define Green's Theorem. Evaluate  $\oint_C \vec{F} \cdot d\vec{r}$ , where  $\vec{F} = (\sin y, \cos x)$  and  $C$  is the triangle with vertices  $(0, 0)$ ,  $(\pi, 0)$ ,  $(\pi, 1)$ , by using Green's theorem. 8
- b) Evaluate  $\oint_C \vec{F} \cdot d\vec{r}$  by using Stoke's theorem, where  $\vec{F} = (y^2, z^2, x^2)$  and  $C$  is the boundary of the surface  $S: x + y + z = 1$  in the first octant. 7
- OR**
- Evaluate the surface integral  $\iint_S \vec{F} \cdot \vec{n} dA$  where  $\vec{F} = (x^2, y^2, z^2)$  and  $s: \vec{r} = (u \cos v, u \sin v, 3v)$ ,  $0 \leq u \leq 1$ ,  $0 \leq v \leq 2\pi$
6. a) Find the fourier series of  $f(x) = x + |x|$  for  $-\pi < x < \pi$ . 7
- b) Expand the function  $f(x) = x^2$  in the interval  $0 < x < \pi$  in half range Fourier cosine series and show that  $\frac{\pi^2}{6} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$  8
7. Solve Any Two: 2x5
- a) Find the general solution of  $2u_x + 2u_y - u = 0$ .
- b) Derive the one-dimensional traffic flow model using conservation law.
- c) Show that the value under integral sign  $\int_{(4,0,3)}^{(-1,1,2)} [(yz + 1)dx + (xz + 1)dy + (xy + 1)dz]$  is exact and evaluate the integral.

# POKHARA UNIVERSITY

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*Attempt all the questions.*

1. a) Evaluate the integral by reversing the order of integration: 5  

$$\int_0^{\pi} \int_x^{\pi} \left( \frac{\sin y}{y} \right) dy dx$$
  - b) Evaluate the integral: 5  

$$\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z dz dy dx$$
  - c) Find the volume of the solid whose base is the region in the  $xy$ -plane 5  
that is bounded by the parabola  $y = 4 - x^2$  and the line  $y = 3x$ ,  
while the top of the solid is bounded by the plane  $z = x + 4$ .
  2. a) Solve by using power series method:  $y'' - 4xy' + (4x^2 - 2)y = 0$ . 7
  - b) Solve the Bessel's equation  $x^2 y'' + xy' + (x^2 - n^2)y = 0$ . 8
- OR
- i. Express  $x^2 - 4x + 2$  as a Legendre polynomial by using Rodrigues formula.
  - ii. Sketch the graph of  $P_0(x)$ ,  $P_1(x)$ ,  $P_2(x)$  and  $P_3(x)$  in a single graph.
3. a) Solve the initial value problem by using Laplace transform: 7  
 $y'' + 2y' + 2y = 5\sin t, y(0) = y'(0) = 0$
  - b) i. Find the Laplace transform of:  $\cosh at \sin at$ . 8  
ii. Find the inverse Laplace transform of  $\frac{s}{(s^2+1)(s^2+4)}$  using the convolution theorem.
4. a) If  $\phi = x^3 + y^3 + z^3 - 3xyz$ , find:  $\text{div}(\text{grad} \phi)$  5
  - b) Find the directional derivative of  $\phi = x^2 - y^2 + 2z^2$  at the point 5  
 $P(1, 2, 3)$  in the direction of line  $PQ$ , where  $Q$  is the point  $(5, 0, 4)$ .
  - c) Find the work done by the force  $\vec{F} = (2y+3)\vec{i} + xz\vec{j} + (yz-x)\vec{k}$  when a 5  
particle moves from  $(0, 0, 0)$  to the point  $(2, 1, 1)$  along the curve  
 $x = 2t^2, y = t, z = t^3$ .

5. a) Find the flux of  $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$  over the portion of the surface  $x^2 + y^2 = 1$  in the first octant between  $z = 0$  to  $z = 2$ . 7
- b) State Stoke's theorem. Use it to evaluate  $\oint_C \vec{F} \cdot d\vec{r}$ , where  $\vec{F} = (z, x, y)$  and  $C$  is the surface of upper half of sphere  $x^2 + y^2 + z^2 = a^2$  in  $xy$ -plane. 8

OR

State Gauss divergence theorem. Evaluate  $\iiint_S \vec{F} \cdot \vec{n} \, ds$ , where  $\vec{F} = (4x, x^2y, -x^2z)$  and  $S$  is the surface of the tetrahedron with vertices  $(0,0,0)$ ,  $(1,0,0)$ ,  $(0,1,0)$ , and  $(0,0,1)$ , by using the Gauss divergence theorem.

6. a) Find the Fourier series of the function  $f(x) = |x|$  for  $-2 < x < 2$ . 7
- b) Expand the function  $f(x) = x^2$  in the interval  $0 < x < \pi$  in half-range Fourier cosine series and deduce that  $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$  8
7. Solve: (Any two) 2×5
- a) Find the general solution of  $u_x + u_y - u = 0$ .
- b) Derive one dimensional traffic flow model using convection law.
- c) Show that the integral  $\int_{(0,0,0)}^{(2,4,0)} e^{x-y+z^2} (dx - dy + 2zdz)$  is exact and evaluate the exact value.